

PAPER_226: SGR 0501+4516 Magnetar — 11-Term Full MUGE with GW Back-Reaction, Magnetic Energy, and Cumulative Burst Decay

Author: Daniel T. Murphy

Session: 58 (grokshare8d951e12.txt extraction — Doc 2)

PAPER_226: SGR 0501+4516 Magnetar — 11-Term Full MUGE with GW Back-Reaction, Magnetic Energy, and Cumulative Burst Decay

Author: Daniel T. Murphy **Framework:** UQFF v4.8 (Star-Magic) **Session:** 58 (grokshare8d951e12.txt extraction — Doc 2) **Date:** March 2026 **Classification:** Novel Magnetar MUGE Physics — Three Uniquely Rare Mathematical Discoveries **Status:** Proof-Quality Whitepaper

Abstract

This paper presents the complete 11-term MUGE (Modified Unified Gravitational Equation) for SGR 0501+4516, a soft gamma repeater magnetar at ~ 2 kpc. Three novel mathematical discoveries are documented: (1) gravitational wave spin-down back-reaction $a_{\text{GW}} = (G \cdot M^2)/(c^4 r) \cdot (d\Omega/dt)^2$, (2) magnetic stored energy acceleration $a_{\text{mag}} = B^2 \sqrt{2\mu_0 M r}$, and (3) cumulative burst-decay energy $a_{\text{decay}} = L_0 \tau_d (1 - e^{-t/\tau_d})/(Mr)$. The computed canonical gravity $g \approx 4.474 \times 10^{12} \text{ m/s}^2$ at $t=5000$ yr is consistent with magnetar surface gravity expectations.

1. Physical System

SGR 0501+4516 is a soft gamma repeater with:

Inferred dipole field $B_0 = 10^{10} \text{ T}$, decaying on $\tau_B = 4000 \text{ yr}$

Rotation period $P = 5.0 \text{ s}$, spin-down on $\tau_\Omega = 10^6 \text{ yr}$

Mass $M = 1.4 M_\odot$, neutron star radius $r = 20 \text{ km}$

Quiescent X-ray luminosity $L_0 \approx 10^{28} \text{ W}$

2. Novel MUGE Terms

2.1 Gravitational Wave Spin-Down Back-Reaction (Novel)

$$a_{\text{GW}} = \frac{G M^2}{c^4 r} \left(\frac{d\Omega}{dt} \right)^2$$

With $d\Omega/dt = -(2\pi/P)/\tau_\Omega \cdot e^{-t/\tau_\Omega}$, this captures the gravitational wave back-reaction torque on the spinning magnetar. This term is unique to SGR 0501+4516 in the MUGE catalogue — no other system uses GW spin-down as a direct acceleration term.

2.2 Magnetic Stored Energy Acceleration (Novel)

$$a_{\text{mag}} = \frac{B(t)^2}{2\mu_0} \cdot \frac{4\pi r^3}{3} \cdot \frac{1}{M \cdot r}$$

The magnetic energy density integrated over the neutron star volume divided by M_r gives an effective acceleration from the stored field energy. This is distinct from the magnetic suppression factor $f_{\text{sc}} = 1 - B/B_{\text{crit}}$ used in other magnetar systems.

2.3 Cumulative Burst Decay Energy (Novel)

$$a_{\text{decay}} = \frac{L_0 \tau_d (1 - e^{-t/\tau_d})}{M_r}$$

This integrates the total burst luminosity energy released up to time t , converting cumulative photon energy to an effective acceleration. At large t : saturates to $L_0 \tau_d / (M_r)$.

3. Full 11-Term MUGE

$$g_{0501} = \underbrace{a_{\text{grav}}}_{\text{rm base}} + \underbrace{a_{\text{Ug}}}_{\text{rm UQFF}} + \underbrace{a_{\Lambda}}_{\text{rm cosm}} + \underbrace{a_{\text{EM}}}_{\text{rm vac}} + \underbrace{a_{\text{GW}}}_{\text{rm spin}} + \underbrace{a_{\text{q}}}_{\text{rm quantum}} + \underbrace{a_{\text{f}}}_{\text{rm fluid}} + \underbrace{a_{\text{osc}}}_{\text{rm osc}} + \underbrace{a_{\text{DM}}}_{\text{rm DM}} + \underbrace{a_{\text{mag}}}_{\text{rm Bmag}} + \underbrace{a_{\text{decay}}}_{\text{rm burst}}$$

At $t = 5000 \text{ yr}$: $g_{0501} \approx 4.474 \times 10^{12} \text{ m/s}^2$.

4. Simulation Parameters

Parameter	Value
M	$1.4 M_{\odot} = 2.785 \times 10^{30} \text{ kg}$
r	$20 \text{ km} = 2 \times 10^4 \text{ m}$
B_0	10^{10} T
τ_B	4000 yr
P_{init}	5.0 s
τ_{Ω}	$10\{, \}000 \text{ yr}$
L_0	10^{28} W
τ_d	1000 s
Δt timestep	1 yr
$t_{\text{canonical}}$	5000 yr

5. Calculator Class

```
class MagnetarSGR0501MUGEFullCalculator(_CP3Calculator):
    """PAPER_226: SGR 0501+4516 – 11-term MUGE with GW back-reaction, mag energy, burst decay"""
    # Session 58 – grok_share_8d951e12.txt
```

Located in CondensedPhysics3.py (Session 58).

6. Conclusion

Three previously undocumented MUGE terms are established for magnetar environments: (1) GW back-reaction spin-down coupling $(G M^2 / c^4 r)(d\Omega/dt)^2$, (2) magnetic energy density acceleration $B^2 V / (2\mu_0 Mr)$, and (3) cumulative burst energy-to-acceleration conversion $L_0 \tau_d (1 - e^{-t/\tau_d}) / (Mr)$. These complete the 11-term SGR 0501+4516 MUGE, the most term-rich magnetar model in the UQFF library.

Source: grokshare8d951e12.txt — Doc 2 (SGR 0501+4516 MUGE Full)

UQFF computed: Eddington luminosity UQFF correction = $1 - [SSq] \exp(-\tau_d t) = 1 - 5.7e-1 \exp(-2.9e-4) = 4.3e-1$; F_U at event horizon = $2.0e+18$ m/s.